MINJUN LONG

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EDUCATION

UNIVERSITY OF VIRGINIA Charlottesville, VA

PhD Candidate, Computer Science (GPA: 4.0/4.0) 2022 – Fall 2027 (Expected)

Bachelor of Arts in Computer Science (GPA: 3.95/4.0), Psychology (GPA 3.82/4.0) 2018 – 2022

RESEARCH INTEREST

My research focuses on systematic evaluation and empirical analysis of security- and privacy-critical systems, with an emphasis on understanding when claimed guarantees break down in real-world settings, from web tracking and advertising privacy protections to privacy-regulation compliance and deployed detection systems. I build threat models, measurement pipelines, and evaluation frameworks to analyze how privacy protections and security constraints hold up across real-world deployments. I am extending this agenda toward AI and agent security, including safeguard evolution, behavioral monitoring, and scalable evaluation of adaptive AI systems.

PUBLICATIONS

IEEE S&P 2027: Long, M., & Evans, D. "Evolution of Log-Based Detection Rules in Public Repositories". To appear.

PETS 2024: Long, M., & Evans, D. "Evaluating Google's Protected Audience Protocol". 24th Privacy Enhancing Technologies Symposium.

PETS 2023: Shezan, F. H., Long, M., Hasani, D., Wang, G., & Tian, Y. "SenRev: Measurement of Personal Information Disclosure in Online Health Communities". 23rd Privacy Enhancing Technologies Symposium.

ACM WPES 2022: Rahat, T. A., Long, M., & Tian, Y. (2022, November). "Is Your Policy Compliant? A Deep Learning-Based Empirical Study of Privacy Policies' Compliance with GDPR". 21st Workshop on Privacy in the Electronic Society.

RESEARCH EXPERIENCE

Agent Security, Permission Boundaries, and Evaluation 2026 – Present

- Designing a policy enforcement layer between autonomous AI coding agents and the external services they act on (e.g., databases, version control).
- Building prototype gateways and live-agent experiments to evaluate enforcement coverage, task utility, and overhead.

Behavioral Evolution of Detection Systems 2025 – 2026

- Conducted a large-scale longitudinal study of detection rule evolution across thousands of real-world security rules and over 100k revisions from Sigma and Splunk repositories.
- Developed structured intermediate representations to canonicalize detection logic, and scalable analysis pipelines to characterize adaptation patterns, rule reversions, coverage expansion, and operational tradeoffs.

Multimodal Deepfake Detection and Defender Asymmetry 2024 – 2025

- Explored audio-visual deepfake detection (cross-modal synchrony, identity modeling, physiological constraints) and published a research reflection on defender advantage: elena6918.github.io/blog/2025/deepfake/

Evaluating Google's Protected Audience Protocol 2023 – 2024

- Evaluated the unlinkability guarantees of Google's Protected Audience (Privacy Sandbox) ad-auction API using its public specifications and Chrome Origin Trial traffic.
- Showed that its reporting mechanisms let colluding advertisers link users across sites despite differential-privacy noise: with default parameters, two colluding buyers re-identify a visitor among 1M candidates with >99% accuracy, and ~200 buyers enable large-scale surveillance.
- Found covert channels (ad creative URLs, bid/score encoding) that bypass k-anonymity, analyzed countermeasures, and shared findings with Google's Protected Audience team.

Automated Large-scale Analysis of Privacy Law Violations 2022 – 2023

- Built data pipelines and deep-learning models to audit real-world websites' privacy policies against GDPR disclosure requirements.
- Processed large-scale web data to extract stated data practices and identify mismatches with regulatory requirements.

OAuth Protocol Vulnerability Analysis 2021 – 2022

- Analyzed OpenID Connect protocols to identify vulnerabilities in open-source implementations.

Measurement of Personal Information Disclosure in Online Health Communities 2020 – 2021

- Built Python crawlers to collect multi-modal, multi-lingual user content (text and medical images) from four online health communities across web and mobile apps.
- Built an automated system to analyze ~1.9M data elements, finding sensitive personal-information leakage in ~70%.

Privacy-Preserving Image Processing 2019 – 2020

- Built a video encryption system that detects and encrypts faces, with eigenface-derived biometric keys so only a matching face can be decrypted.

TECHNICAL SKILLS

Programming Languages & Tools: Python, SQL, JavaScript, Linux, Git, AWS

Security & Systems: Privacy/security measurement, web tracking, threat modeling, detection engineering, GDPR auditing

AI/ML & Evaluation: Machine learning pipelines, LLM-assisted analysis, agent evaluation, behavioral analysis, longitudinal/statistical evaluation

INDUSTRY EXPERIENCE

Anhui Wantong Technology Hunan, China

Software Engineer Intern 2019 Summer

- Designed a MySQL database and developed web front-end and back-end monitoring modules (Spring MVC) for the Hunan Public Security Department.

ACADEMIC SERVICE

Artifact Reviewer: PETS 2025-2027, SIGCOMM 2025, USENIX Security 2026

External Peer Reviewer: IEEE IoT-J 2024, IEEE TDSC 2024, IEEE TDSC 2025